

Flag 5 (Exported Broadcast Receiver)

Analysis & Exploitation Steps:-

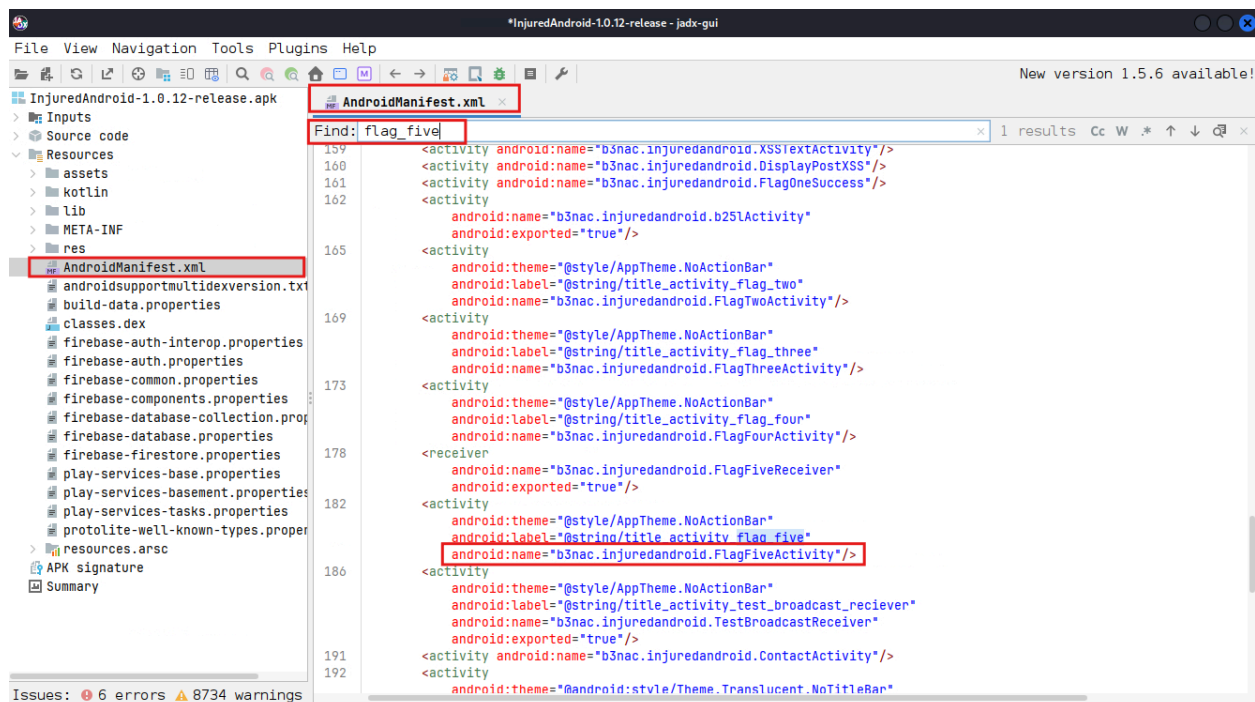


Figure 1: Analyzing AndroidManifest.xml to Outline of Everything the Application Does and Searching for the String "flag_five" and Found an Activity of Class "b3nac.injuredandroid.FlagFiveActivity"

```

FlagFiveActivity x
40     F();
41 }
42
43 @Override // androidx.appcompat.app.c, androidx.fragment.app.d, androidx.activity.ComponentActivity, androidx.core
44 protected void onCreate(Bundle bundle) {
45     super.onCreate(bundle);
46     setContentView(R.layout.activity_flag_five);
47     C((Toolbar) findViewById(R.id.toolbar));
48     ((FloatingActionButton) findViewById(R.id.fab)).setOnClickListener(new View.OnClickListener() { // from class:
49         @Override // android.view.View.OnClickListener
50         public final void onClick(View view) {
51             this.f.G(view);
52         }
53     });
54     Button button = (Button) findViewById(R.id.button9);
55     new ComponentName(this, (Class<?>) FlagFiveReceiver.class);
56     getPackageManager();
57     a.m.a.a.b(this.x, new IntentFilter("com.b3nac.injuredandroid.intent.action.CUSTOM_INTENT"));
58     button.setOnClickListener(new View.OnClickListener() { // from class: b3nac.injuredandroid.b
59         @Override // android.view.View.OnClickListener
60         public final void onClick(View view) {
61             this.f.H(view);
62         }
63     });
64 }
65
66 @Override // androidx.appcompat.app.c, androidx.fragment.app.d, android.app.Activity
67 protected void onDestroy() {
68     a.m.a.a.b(this).e(this.x);
69     super.onDestroy();
70 }
71 }

```

Figure 2: Displays the Decompiled Source for "b3nac.injuredandroid.FlagFiveActivity" Class

```

FlagFiveReceiver x
16     public void onReceive(Context context, Intent intent) {
17         String string;
18         int i;
19         d.s.d.g.e(context, "context");
20         d.s.d.g.e(intent, "intent");
21         j.j.a(context);
22         int i2 = f1454a;
23         if (i2 != 0) {
24             string = "Keep trying!";
25             if (i2 != 1) {
26                 if (i2 != 2) {
27                     Toast.makeText(context, "Keep trying!", 1).show();
28                     return;
29                 }
30                 String str = "You are a winner " + k.a("Zkdlt0WwtLQ=");
31                 new FlagsOverview().H(true);
32                 new j().b(context, "flagFiveButtonColor", true);
33                 Toast.makeText(context, str, 1).show();
34                 i = 0;
35             }
36             f1454a = i;
37         }
38         StringBuilder sb = new StringBuilder();
39         sb.append(d.w.h.e("\n    Action: " + intent.getAction() + "\n\n    "));
40         sb.append(d.w.h.e("\n    URI: " + intent.toUri(1) + "\n\n    "));
41         string = sb.toString();
42         d.s.d.g.d(string, "sb.toString()");
43         Log.d("DUUDE!", string);
44         Toast.makeText(context, string, 1).show();
45         i = f1454a + 1;
46         f1454a = i;
47     }

```

Figure 3: Displays the Trigger Logic for BroadcastReceiver from the Exported Receiver

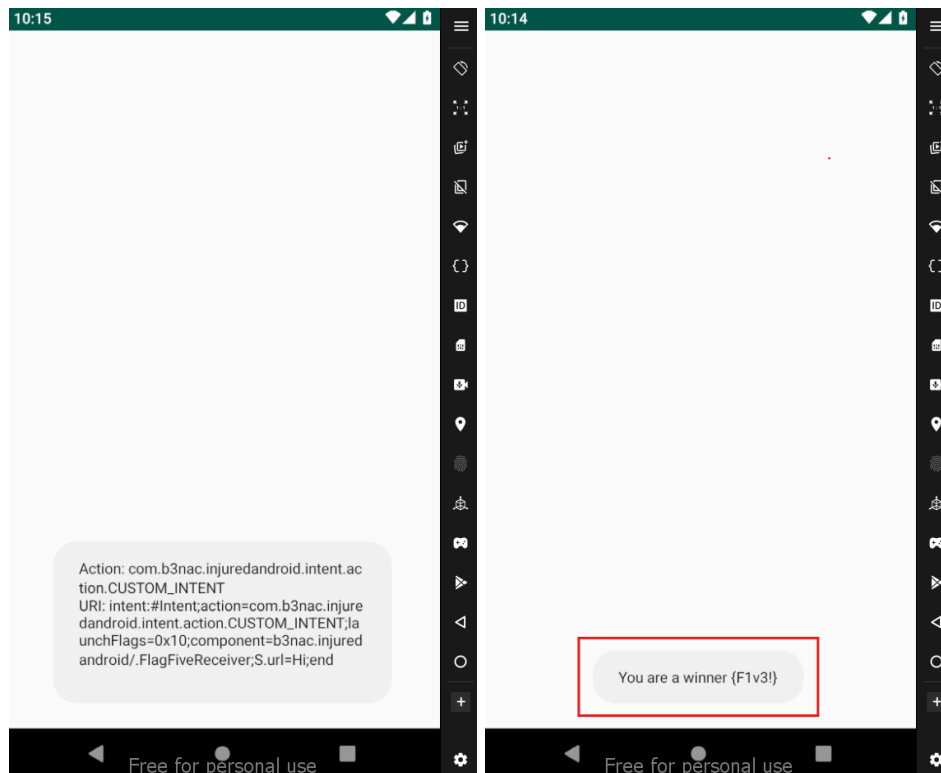


Figure 4: Displays Successful Submission of the Flag After Opening “FLAG FIVE - EXPORTED BROADCAST RECEIVER” a Couple of Times to Trigger It

Impacts:-

- **Exported BroadcastReceiver allows unauthorized apps to trigger sensitive functionality & expose it to an attacker**

Remediations:-

- **Set android:exported="false" when external access is unnecessary, use permissions and validate incoming intents when access is required & avoid exposing sensitive data through broadcasts**